

PRODUCT REQUIREMENTS DOCUMENT

GeoAtlas

1. Product Vision

Create a **real-time intelligence platform** that connects **global geopolitical events** with **financial market movements** and **predictive insights**.

Users should be able to understand:

```
Global Event
  ↓
Sector Impact
  ↓
Market Movement
  ↓
Predicted Outcome
```

Example:

Event:
China sanctions Taiwan semiconductor exports

Impact:
Semiconductor supply chain risk

Affected Assets:
TSMC
NVIDIA
AMD

Prediction:
Short term volatility ↑
Chip ETFs ↓

2. Problem Statement

Current platforms separate **news**, **geopolitics**, and **markets**.

Platform	Problem
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Google News	No market context
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TradingView	No geopolitical analysis
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Bloomberg	Expensive
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Platform	Problem
Twitter/X	No signal filtering

Users cannot quickly answer:

“How will this global event affect markets?”

3. Target Users

Primary

Retail traders
macro investors
geopolitics enthusiasts

Secondary

- hedge fund analysts
- policy researchers
- journalists
- students

4. Unique Value Proposition

The first platform that maps geopolitical events to market movements and AI predictions in real time.

Key differentiation:

Feature	Traditional Platforms	GeoAlpha
News	Raw feed	Structured event intelligence
Market data	Price charts	Event-driven analysis
Prediction	None	AI prediction
Knowledge organization	None	Pinterest-style boards

5. Core Product Modules

Module 1 — Global Event Intelligence Feed

A real-time feed of geopolitical and macroeconomic events.

Example feed card:

Event:
Russia attacks Ukrainian refinery

Category:
Conflict

Impact:
Oil supply disruption

Affected Markets:
Crude Oil
Energy Stocks

Prediction:
Oil ↑ short term

Sources

- Reuters
- AP News
- Al Jazeera
- Government releases
- Think tank reports

Data Sources APIs

GDELT
NewsAPI
Mediastack
EventRegistry

Module 2 — Event Intelligence Engine

Convert news into **structured geopolitical events**.

Pipeline:

News Article
↓
NLP Extraction
↓
Event Classification
↓
Sector Impact
↓
Market Mapping

Example structured output:

```
{
  event: "China sanctions US chip firms",
  country: "China",
  sector: "Semiconductors",
  impact: "negative",
  assets: ["NVDA", "AMD", "TSMC"],
  confidence: 0.72
}
```

Models Used

- Named Entity Recognition
- Event extraction
- Sentiment analysis
- Topic classification

Module 3 — Live Market Data Engine

Integrates global market data.

Assets supported:

- Stocks
- ETFs
- Commodities
- Crypto
- Forex

Data APIs:

Polygon
Alpha Vantage
Finnhub
TwelveData

Features:

- real-time price
- volume
- charts
- volatility

Module 4 — AI Market Impact Prediction

Predict **probable market reaction to geopolitical events.**

Example:

Event:

Iran blocks Strait of Hormuz

Prediction:

Oil price:

↑ 12% probability

Shipping stocks:

↓ 8%

Confidence:

65%

Models used:

ML models

LSTM time series

Transformer sentiment model

Gradient boosting

Inputs:

news sentiment

historical event impact

macro indicators

Module 5 — Intelligence Boards (Pinterest-style)

Users organize intelligence like a **visual knowledge board.**

Example boards:

China-US Tech War

Middle East Conflict

Global Energy Crisis

AI Supply Chain

Users pin:

- news
- market charts
- predictions
- reports

Module 6 — Macro Intelligence Dashboard

High-level geopolitical view.

Features:

- world map event visualization
- heatmaps
- global market trends

Example:

Red regions = conflicts
Green regions = economic growth

Module 7 — Personalized Alerts

Users receive alerts when events affect tracked assets.

Example:

Alert:
China announces export restriction

Affected:
TSMC
NVDA

Prediction:
Short term volatility spike

6. System Architecture

Production-ready architecture.

```
Users
↓
Frontend (Web + Mobile)
↓
API Gateway
↓
Backend Services
↓
AI Processing Layer
↓
Data Pipelines
↓
```

Database

7. Tech Stack (Production Grade)

Frontend (Web)

Next.js
TypeScript
TailwindCSS
ShadCN UI
Recharts
TradingView widgets

Mobile App

React Native
Expo
TypeScript

Single codebase for:

iOS
Android

Backend

FastAPI (Python)
Node.js (microservices)
GraphQL API

AI/ML Layer

PyTorch
HuggingFace Transformers
spaCy
LangChain
scikit-learn

Data Pipeline

Apache Kafka
Apache Airflow
Celery
Redis

Database

PostgreSQL
TimescaleDB (time series)
Redis (caching)

Search

Elasticsearch

Infrastructure

Cloud:

AWS
or
GCP

Services:

Docker
Kubernetes
Terraform

Authentication

Auth0
or
Firebase Auth

8. Data Pipeline

News API ingestion
↓
Article parsing
↓
Event extraction
↓
Sentiment analysis
↓
Market mapping
↓
Prediction engine
↓
Dashboard update

Update cycle:

Every 5-10 minutes

9. Core Database Schema

Users

```
id
email
password
preferences
```

Events

```
event_id
title
country
sector
impact
timestamp
confidence
```

Assets

```
ticker
name
sector
market
```

Event Asset Mapping

```
event_id
ticker
impact
```

Predictions

```
event_id
ticker
prediction
probability
```

Boards

```
board_id
user_id
title
description
```

Pins

```
pin_id  
board_id  
content_type  
content_id
```

10. Novelty / Innovation

The novelty lies in **event-driven market intelligence**.

Most platforms show:

Price → Explanation

This platform does:

Event → Prediction → Market Reaction

AI-driven **event-to-market mapping** is the key innovation.

11. Monetization

Free tier

- basic news
 - limited predictions
 - limited boards
-

Pro (\$12–20/month)

- advanced analytics
 - full predictions
 - unlimited boards
 - alerts
-

Institutional tier

API access
\$100–\$500/month

12. Risk Analysis

Risk 1 — Data licensing

Market APIs can be expensive.

Mitigation:

Use free tier APIs initially

Risk 2 — Prediction accuracy

Bad predictions reduce credibility.

Mitigation:

display probabilities not guarantees

Risk 3 — News bias

Different sources have bias.

Mitigation:

multi-source aggregation

Risk 4 — Scalability

Real-time data pipelines require infrastructure.

Mitigation:

microservices architecture

13. MVP Scope (First Version)

Must include ONLY:

Live geopolitics feed
Event classification
Market data panel
Event → market mapping
Basic prediction model

User boards

Exclude initially:

mobile push alerts
advanced ML models
complex dashboards

14. Development Timeline

Phase 1 — Foundation (Month 1)

Backend
Database
News ingestion
Basic UI

Phase 2 — Intelligence Engine (Month 2)

NLP event extraction
Sector mapping
Market APIs

Phase 3 — Prediction Layer (Month 3)

ML model
Prediction engine
dashboard integration

Phase 4 — Mobile App (Month 4)

React Native app
user boards
notifications

15. Metrics for Success

Key metrics:

DAU (daily active users)
Time spent on platform
Prediction engagement
Board creations

16. Next 3 Immediate Actions

1 Validate product positioning

Decide if you target:

Traders
Geopolitics enthusiasts
Both

2 Choose data sources

Shortlist:

NewsAPI
GDELT
Finnhub

3 Design the MVP architecture

Start with:

Next.js
FastAPI
PostgreSQL
Redis

💡 Important Insight

If executed properly, this project could become:

- a **serious startup idea**
- a **top-tier ML/AI portfolio project**
- a **Bloomberg-style intelligence tool**